

Pandit Deendayal Energy University

Department of Computer Science & Engineering

Machine Learning Laboratory

Lab Assignment 2

Experiment

Household Power Consumption Analysis

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--ALSO UPLOADED THE JUPYTER NOTEBOOK ALONGWITH THIS PDF--

Objective

The purpose of this laboratory assignment is to analyse household electricity consumption data collected at one-minute intervals over nearly four years. The work focuses on understanding the dataset through preprocessing and visualisation rather than predictive modelling. The accompanying Jupyter Notebook contains the complete implementation, code and generated figures.

Dataset Description

The dataset contains minute-wise measurements of household electrical consumption, including Global Active Power, Global Reactive Power, Voltage, Global Intensity and three sub-metering variables. During preprocessing, the Date and Time columns were combined into a single Datetime index, allowing efficient time-series analysis.

Methodology

The dataset was imported into Python using Pandas. After loading the data, the Date and Time columns were merged into a Datetime index and numerical columns were converted to the appropriate data type. Missing values and outliers were examined before extracting the required records for December 2006 and November 2009. One week of November 2009 was additionally used for selected visualisations to improve readability. The notebook submitted alongside this report contains all source code, outputs and visualisations.

Results and Discussion

The generated visualisations provide different perspectives of household electricity consumption. Histograms were used to understand the distribution of power usage, while time-series plots highlighted variations over time. Sub-metering plots compared the energy usage of different household sections. Scatter plots and joint plots were useful for studying relationships between electrical variables. Heatmaps summarised correlations among numerical features, whereas boxplots clearly highlighted the presence of outliers. Bar charts, pie charts and count plots were used to compare average consumption and categorical distributions, making the overall analysis easier to interpret.

Best Visualisation Technique

Each visualisation has a specific purpose. Histograms are useful for analysing distributions, boxplots for detecting outliers and heatmaps for identifying correlations. However, because the dataset represents continuous measurements over time, the time-series plot provides the clearest overall understanding of consumption behaviour. It effectively highlights daily fluctuations, usage peaks and long-term trends that are difficult to observe using other plots.

Conclusion

The dataset was successfully cleaned, explored and analysed using multiple visualisation techniques. Each plot contributed different insights into electricity consumption patterns and the relationships among electrical measurements. Combining these visualisations provides a comprehensive understanding of household energy usage. The accompanying Jupyter Notebook contains the complete implementation, code and generated figures used in this analysis.

References

1. UCI Machine Learning Repository – Individual Household Electric Power Consumption Dataset.
2. Pandas Documentation.
3. Matplotlib Documentation.
4. Seaborn Documentation.